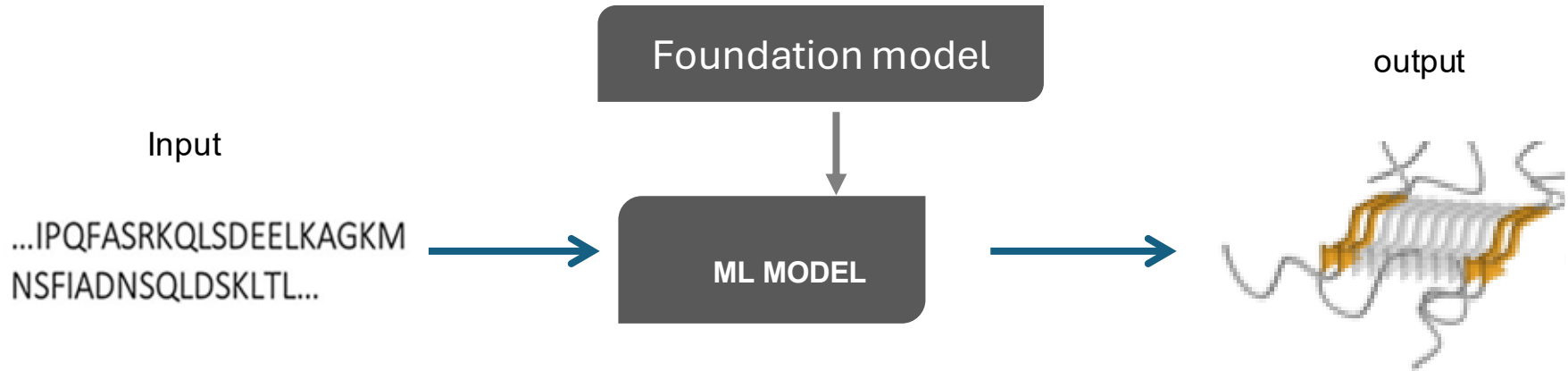




Fine Tuning Biological Foundation Models





Fine Tuning Biological Foundation Models

Jan van Eck



Disclaimer: many slides, made by Jan



What are common difficulties with AI in the Life Science domain?



Go to wooclap.com

Enter the event code in the top banner

Event code

EQRXHUE



Waiting for participants...



Problems specific to AI with Life Science data

a) Life Science data has the wrong dimensions



observations →





Problems specific to AI with Life Science data

Text:

Can you guess what the next word in this sentence ...



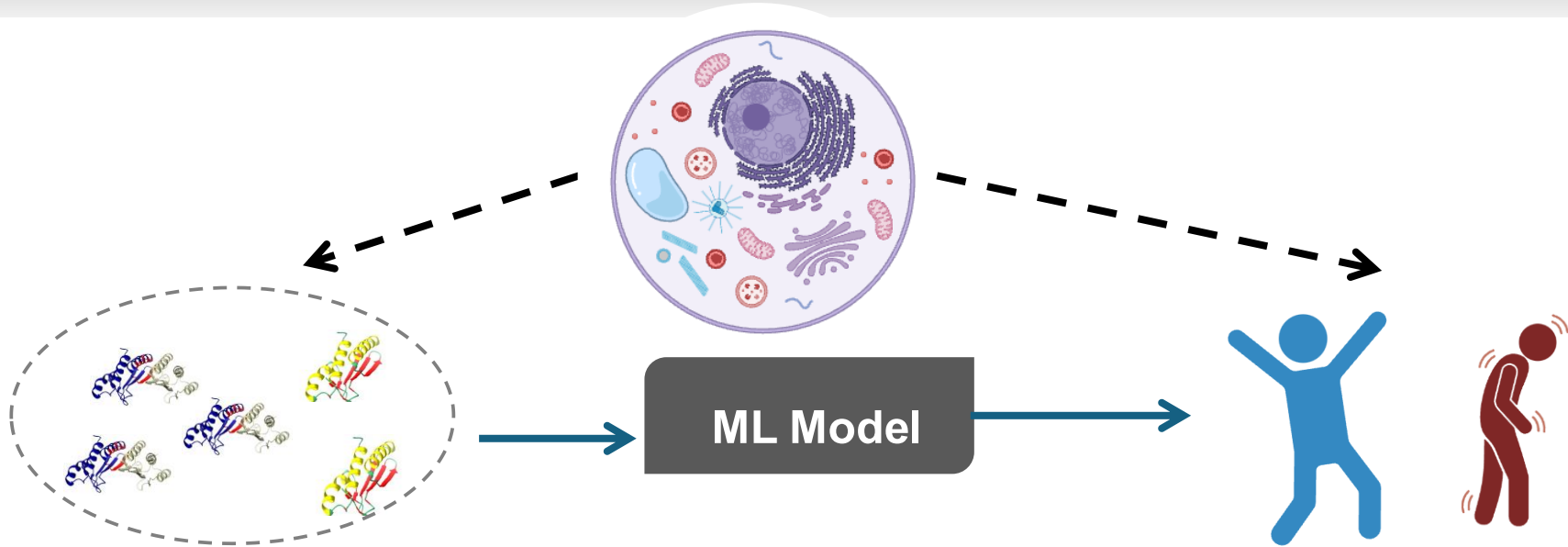
- b) We have no intuitive understanding of the data that we give as input for our models

DNA:





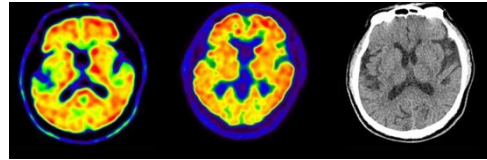
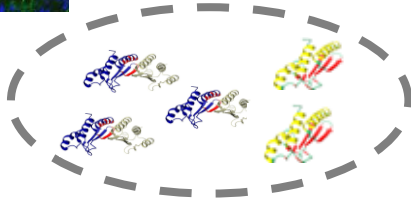
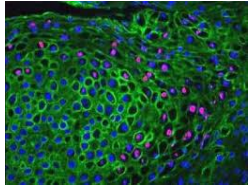
Problems specific to AI with Life Science data



- c) We do not only want to make predictions from the data, but also learn to understand the systems



More problems specific to AI with Life Science data



- Complex, heterogeneous **multi-modal** datasets.
- **Missing data**
- Low **signal-to-noise** ratios.
- Strong to extreme **class imbalance** in most datasets.



AI Technology for Life group

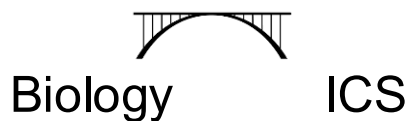


Utrecht
University

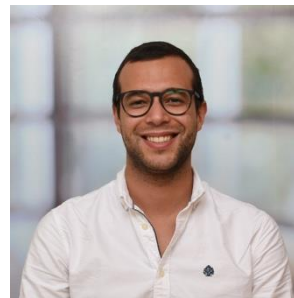
Ronnie de
Jonge



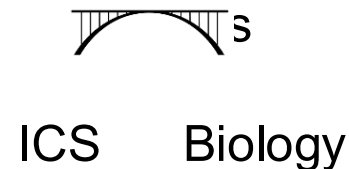
- AI for plant systems
- Plant – Microbiome



Wilson Silva



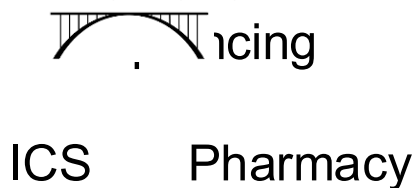
- Explainable AI
- Medical image



Pavel Sinitcyn



- AI for proteomics
- De novo



Sanne Abeln



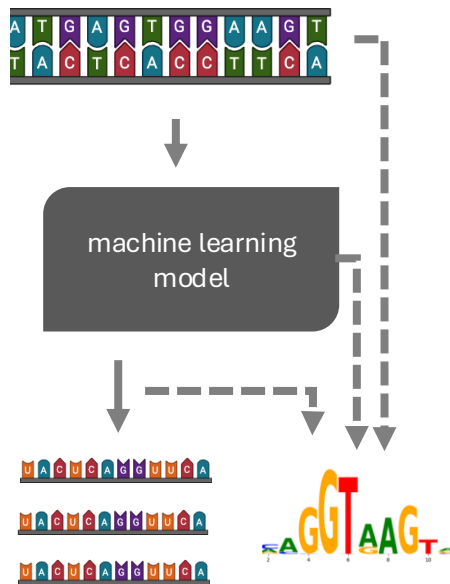
- Explainable AI for multi-omics
- simulations



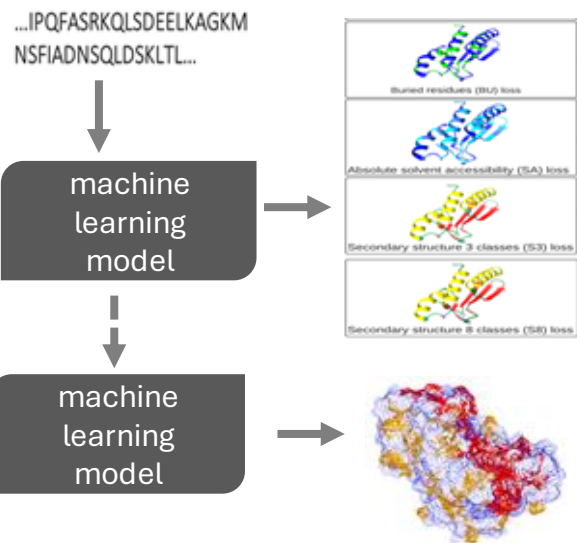


AI Technology for Life group: methodology development

Explainable AI



Transfer learning



Multi-modal data learning

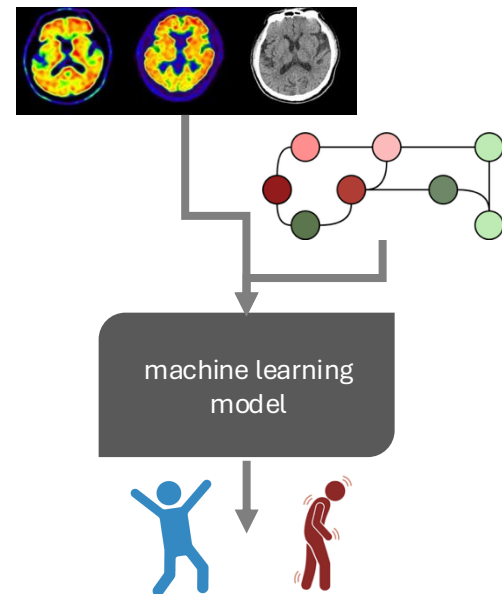




Table of contents

- 01 Biological foundation models and representations
- 02 Frozen embeddings, LoRA and fine tuning protein language models
- 03 Fine-tuning protein language models boosts predictions across diverse tasks



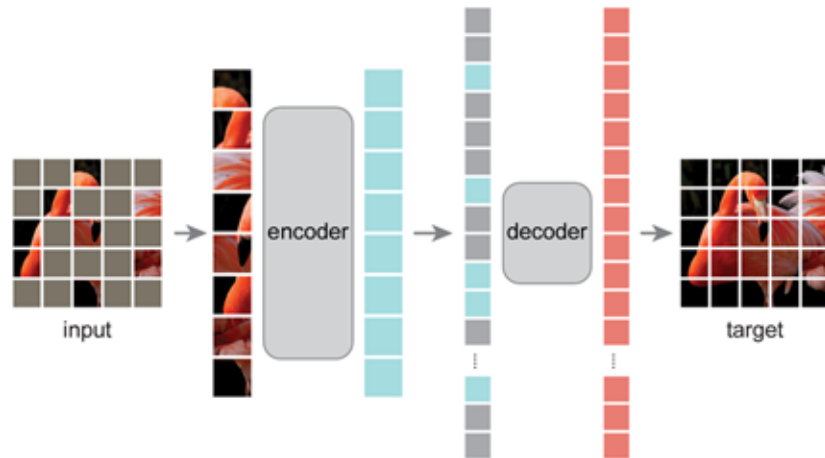
01

Recap and Context



What are foundation models?

- Large neural network trained on broad unlabelled datasets
- Learns general-purpose representations
- Uses self-supervised learning
- Same model can be adapted to many downstream tasks

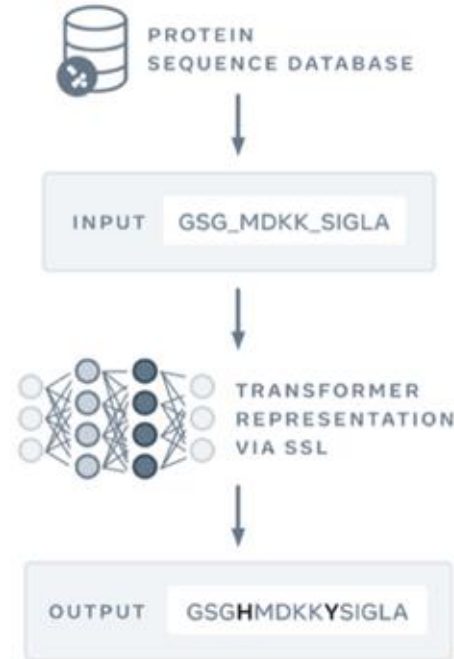




From Foundations Models to Protein Foundation Models

- Protein sequences as input
- Amino acids act like tokens
- Protein language models learn sequence patterns
- Trained on millions to billions of sequences

Pretraining self-supervision on sequences only.



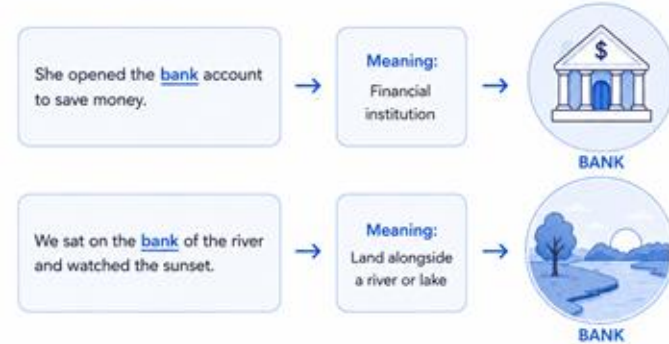


Why Protein Sequences Behave Like Language

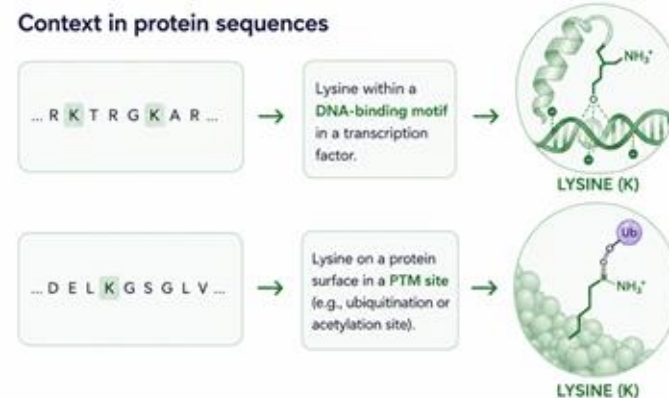
- Amino acid context matters
- Long-range interactions exist
- Sequence order determines function and structure
- Similar sequences often share biological properties



Context in text



Context in protein sequences

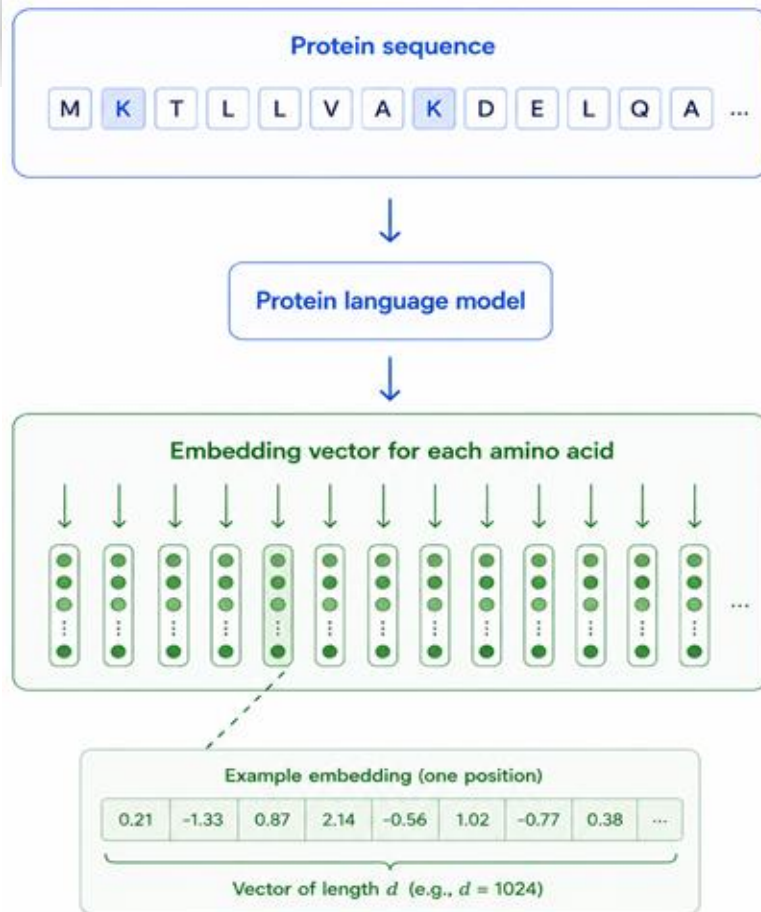




Embeddings as Learned Biological Representations

- Each amino acid position is converted into an embedding vector
- An embedding is simply a vector of learned numerical values
- The embedding depends on the surrounding sequence context

Embeddings as Learned Biological Representations

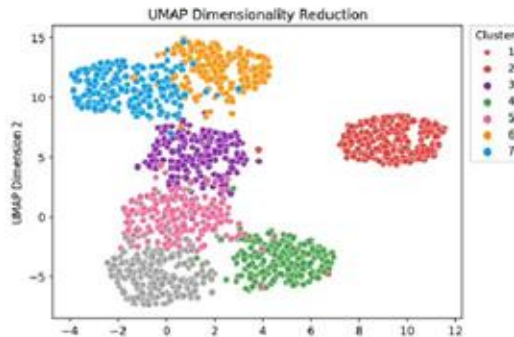
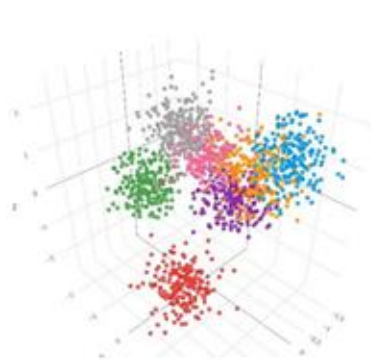




Dimensionality Reduction Techniques

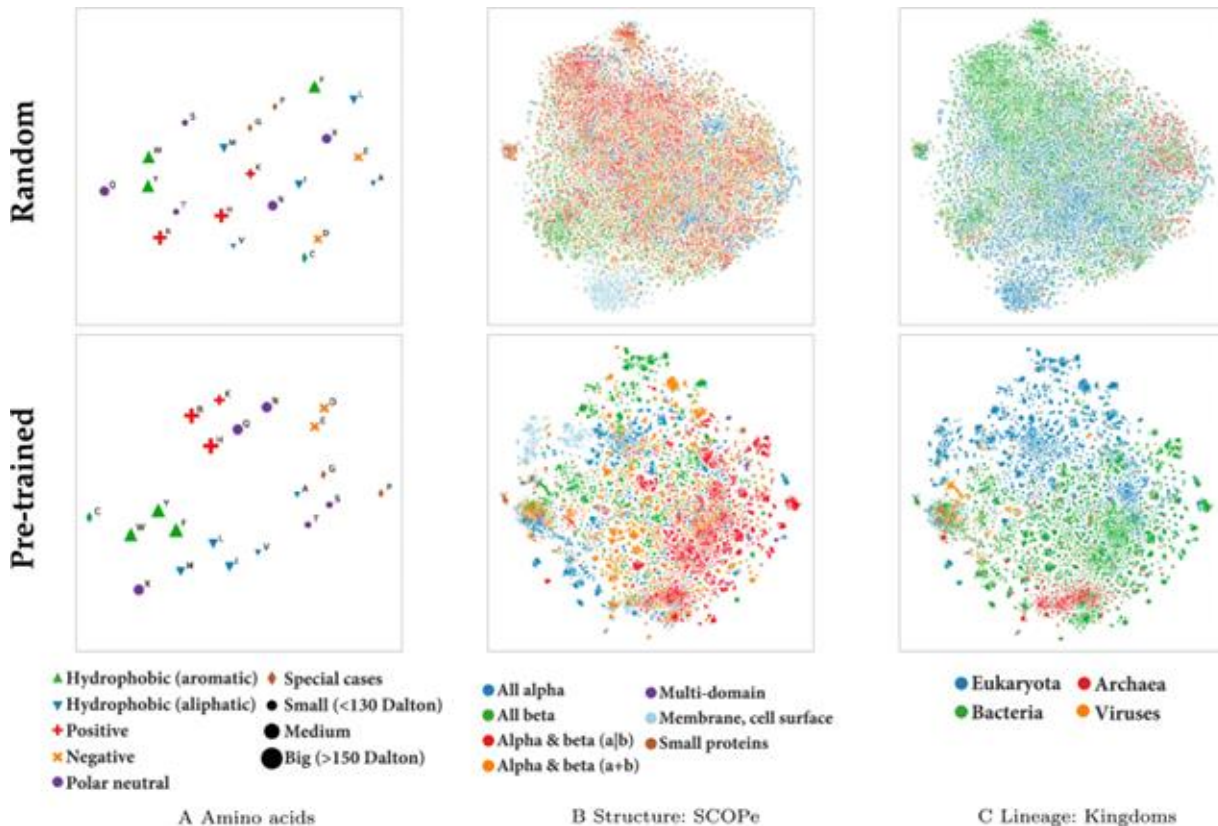
- Preserves local and global structure in high-dimensional data
- Projects complex embeddings into 2D or 3D space for visualization
- Reveals clusters, patterns, and relationships between samples

Point cloud in a 3-dimensional space





What Information Exists Inside Embeddings?





02

Fine Tuning Fundamentals



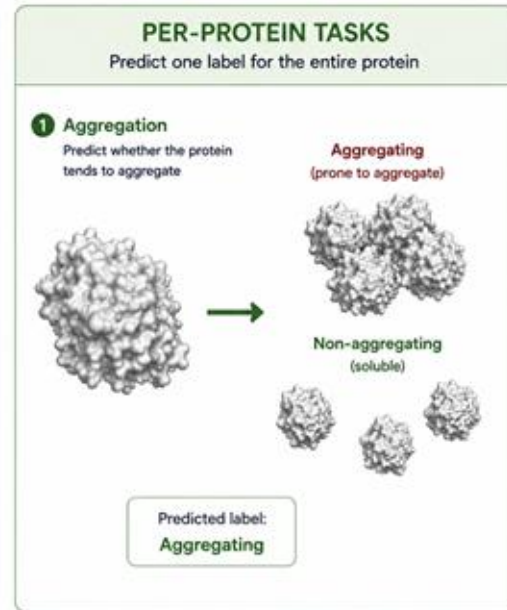
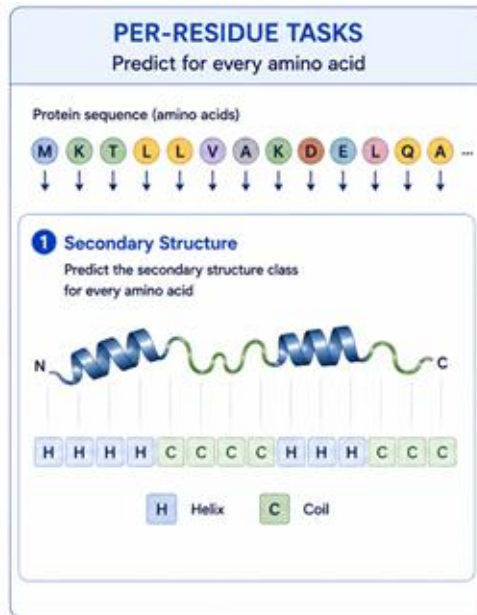
Per-Residue vs Per-Protein Tasks

Predict for every amino acid

- Secondary Structure
- Disorder

Predict one label for entire protein

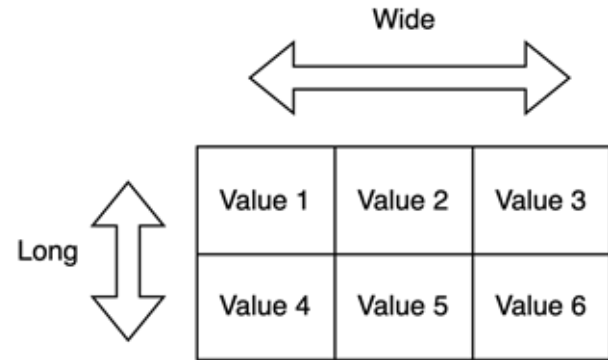
- Localization
- Stability





Small Data Problems in Biology

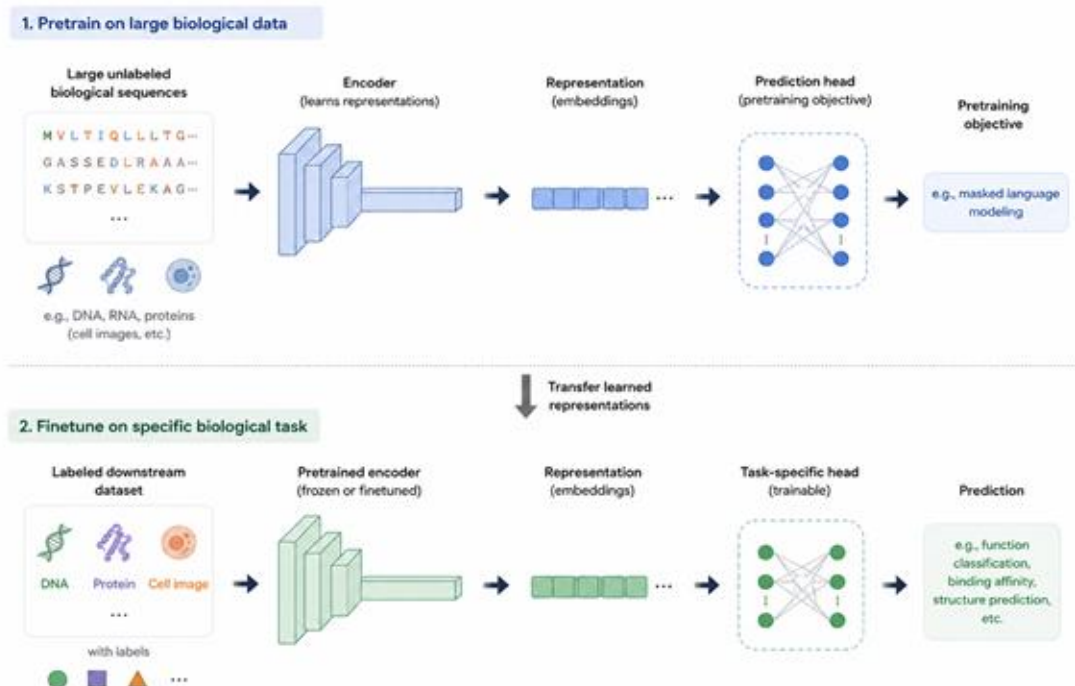
- Biological datasets are often limited in size and highly imbalanced
- Small sample sizes increase overfitting and reduce model generalization
- Transfer learning and foundation models help overcome data scarcity





Transfer Learning

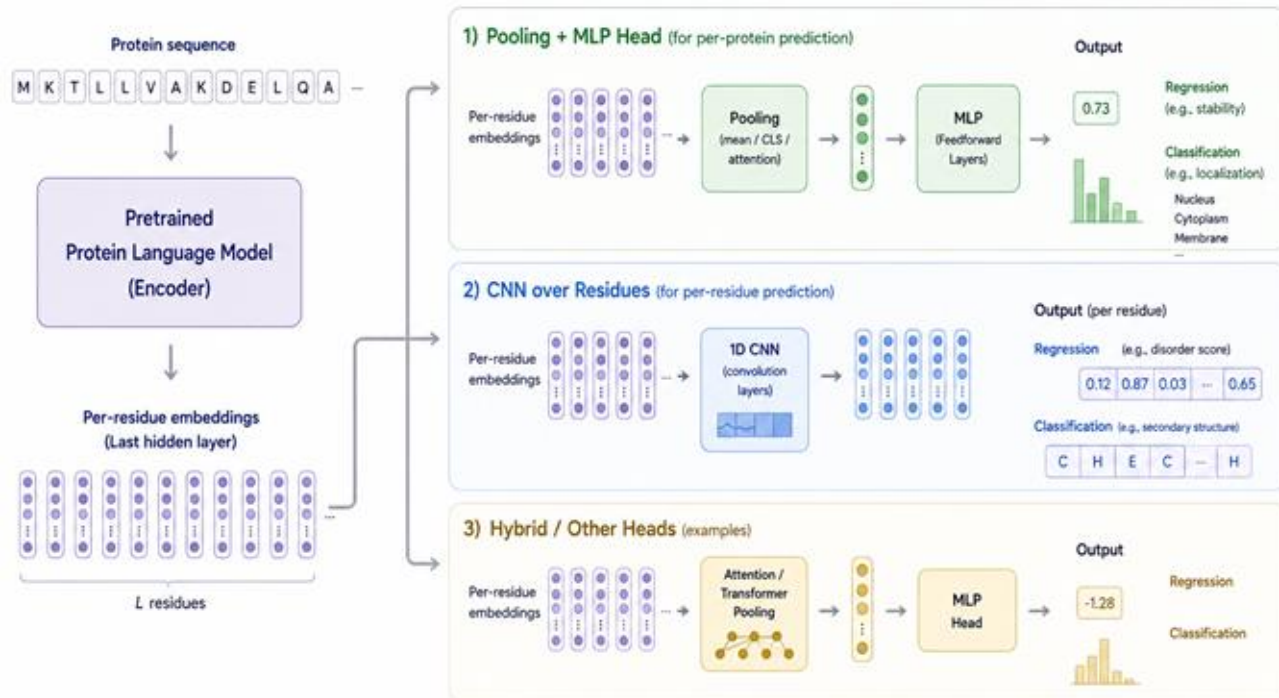
- Pretrain once on massive datasets
- Reuse model for smaller tasks
- Reduces need for expensive labels
- Improves performance on limited datasets





Prediction Heads: final layer(s)

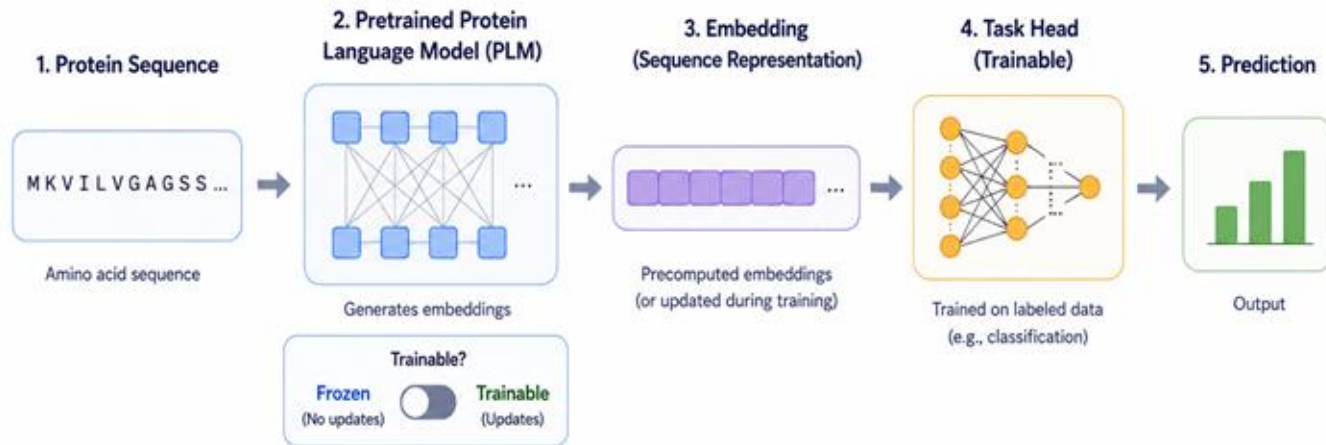
- Linear classifiers
- CNN heads
- Attention pooling
- Regression outputs
- Architecture depends on





Frozen Fine Tuning vs Full Fine Tuning

- Extract embeddings
- Train small classifier/regressor
- Encoder weights of foundation model remain fixed
- Update foundation model weights





Frozen Embeddings

- Computationally cheap
- Stable training
- Requires less GPU memory
- Strong early benchmark performance
- Performs well for many biological tasks

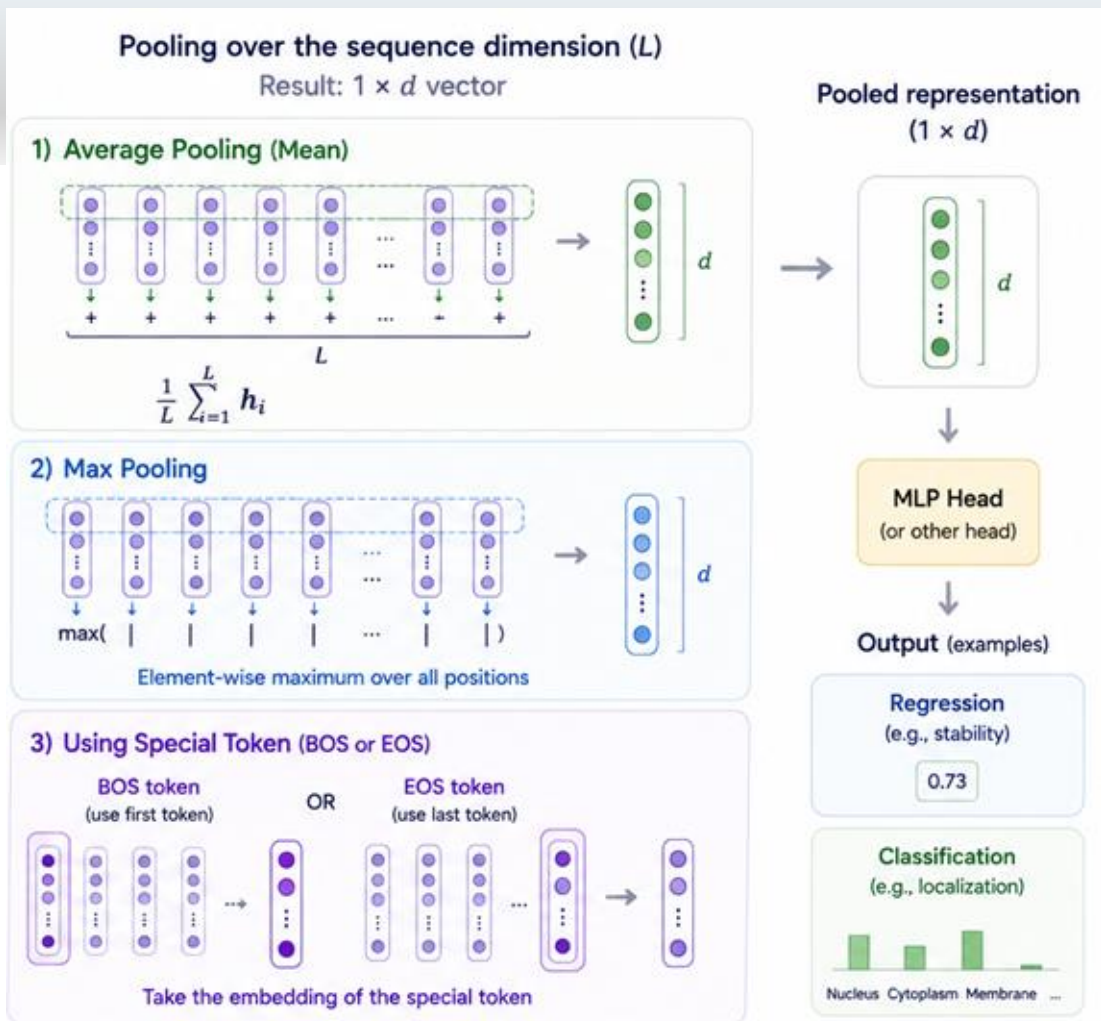
Limitations of Frozen Embeddings

- Representations are static
- Last hidden layer may not be task-optimal
- Information in intermediate layers may be ignored



Pooling Techniques

- Average Pooling
- Max Pooling
- Special Token Pooling





Why Full Fine Tuning Can Improve Performance

- Embeddings are not task-optimal
- Hidden representations can specialize
- Not directly optimized for biology tasks
- Middle layers may contain richer information
- Fine-tuning unlocks these layers

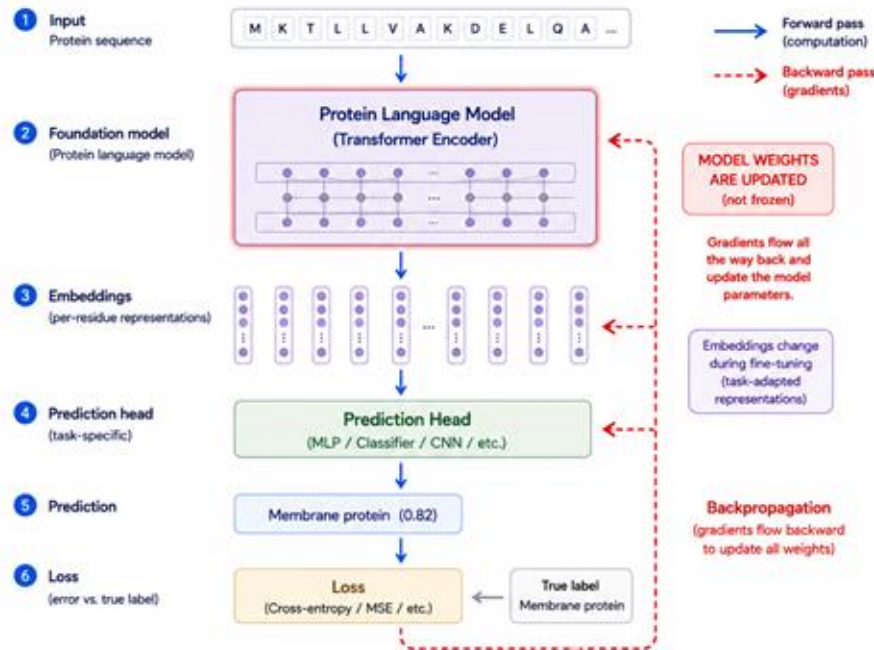
Limitations of Full Fine Tuning

- Model can lose pretrained knowledge
- Especially dangerous with small datasets
- Overfitting Risks in Biology
- Complex models memorize easily
- Need careful benchmarking



Backpropagation Through the Foundation Model

- Update Classification Head
- Update Foundation Model
- Hidden Representations Change





Computational Challenges of Fine-Tuning



- Large GPU memory requirements
- Long sequences increase cost
- Training billion-parameter models is expensive
- Compute becomes a bottleneck

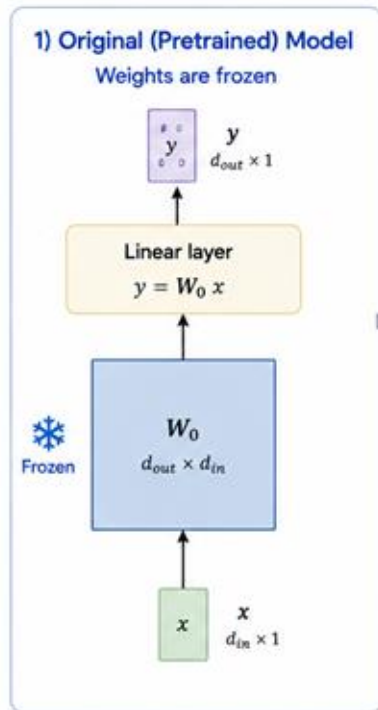


Parameter-Efficient Fine-Tuning (PEFT)

- Only update small subsets of parameters
- Freeze most weights
- Reduce memory usage
- Faster training
- Popular for LLMs and PLMs

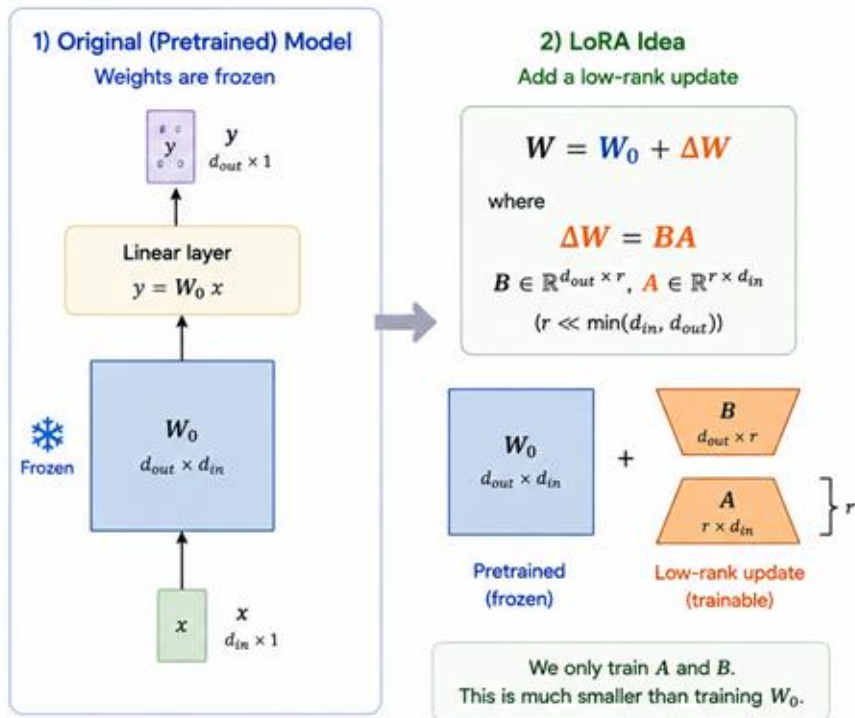


How Lora Works





How Lora Works



Example:

$$R^{d_{out}}, R^{d_{in}} = 1280$$

$$r = 8$$

Trainable weights full model:

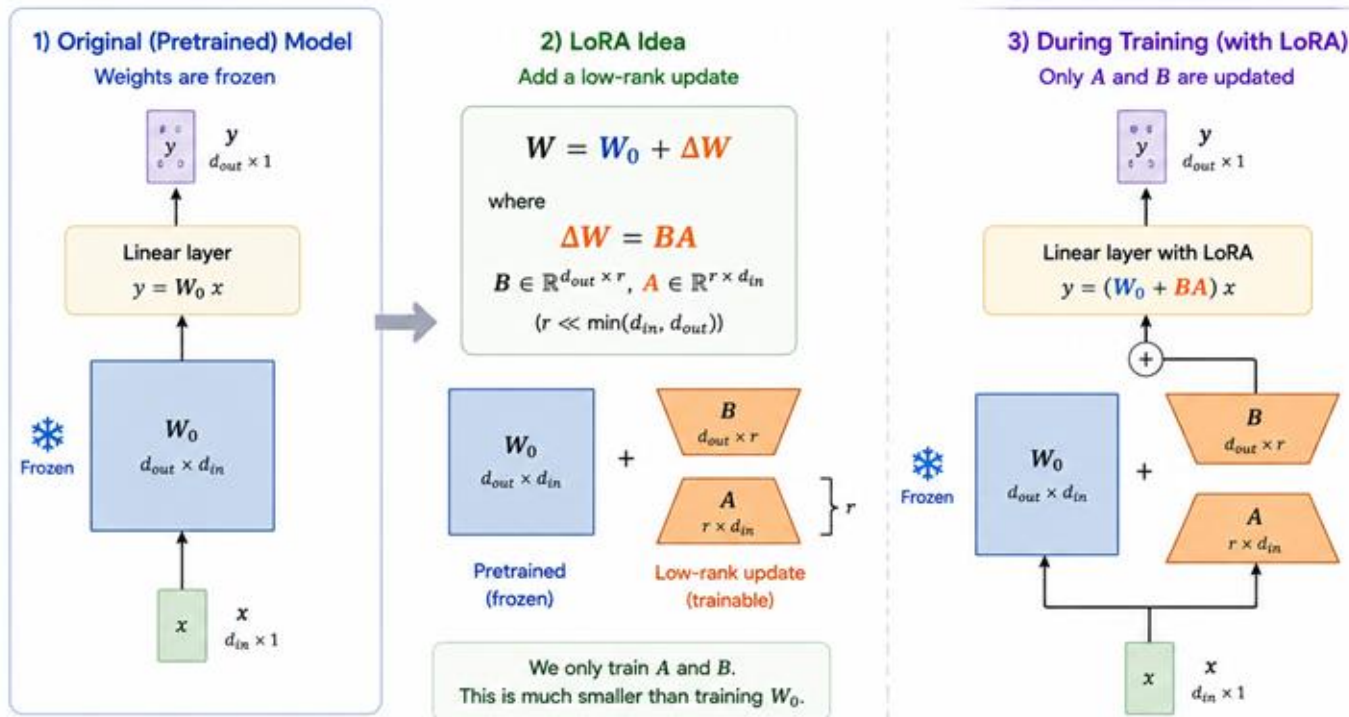
- $1280 \times 1280 = 1,638,400$

Trainable weights LoRa:

- $1280 \times 8 + 8 \times 1280 = 2,0480$



How Lora (Low-rank adaptation) Works





LoRA vs Full Fine-Tuning

- LoRA is faster
- LoRA uses less memory
- Performance often comparable
- Tradeoff depends on task and scale



03

Testing different fine tuning strategies

Fine-tuning protein language models boosts predictions across diverse tasks



Fine-tuning protein language models boosts predictions across diverse tasks

Received: 25 January 2024

Accepted: 15 August 2024

Published online: 28 August 2024

 Check for updates

Robert Schmirler ^{1,2,3} , Michael Heinzinger ¹ & Burkhard Rost ^{1,4,5}

Prediction methods inputting embeddings from protein language models have reached or even surpassed state-of-the-art performance on many protein prediction tasks. In natural language processing fine-tuning large language models has become the de facto standard. In contrast, most protein language model-based protein predictions do not back-propagate to the language model. Here, we compare the fine-tuning of three state-of-the-art models (ESM2, ProtT5, Ankh) on eight different tasks. Two results stand out. Firstly, task-specific supervised fine-tuning almost always improves downstream predictions. Secondly, parameter-efficient fine-tuning can reach similar improvements consuming substantially fewer resources at up to 4.5-fold acceleration of training over fine-tuning full models. Our results suggest to always try fine-tuning, in particular for problems with small datasets, such as for fitness landscape predictions of a single protein. For ease of adaptability, we provide easy-to-use notebooks to fine-tune all models used during this work for per-protein (pooling) and per-residue prediction tasks.



Research Question

- Does fine-tuning improve protein predictions?
- Can PEFT match full fine-tuning?
- Which tasks benefit most?
- What are the compute tradeoffs?
- Can small datasets benefit?



Models Compared in the Study

- ESM2 family
- ProtT5
- Ankh

Table 1 | Protein language models (pLMs) applied in the study*

Model	Architecture (pretraining)	Number of parameters (encoder)	Trained parameters LoRA	Encoder layers	Emb size	Huggingface model checkpoint
Ankh Base	Encoder-Decoder	736 M	2100 K	48	768	ankh-base
Ankh Large		1900 M	4900 K	48	1536	ankh-large
ProtT5		1200 M	3500 K	24	1024	prot_t5_xl_uniref50
ProstT5		1200 M	3500 K	24	1024	ProstT5
ESM2 8 M	Encoder	8 M	163 K	6	320	esm2_t6_8M_UR50D
ESM2 35 M		35 M	483 K	12	480	esm2_t12_35M_UR50D
ESM2 150 M		150 M	1600 K	30	640	esm2_t30_150M_UR50D
ESM2 650 M		650 M	3500 K	33	1280	esm2_t33_650M_UR50D
ESM2 3B		3000 M	7700 K	36	2560	esm2_t36_3B_UR50D

*Emb size provides the dimension of the embeddings of the corresponding pLM. Throughout the paper, we used the standard acronyms K for 10^3 , M for 10^6 , and G for 10^9 .



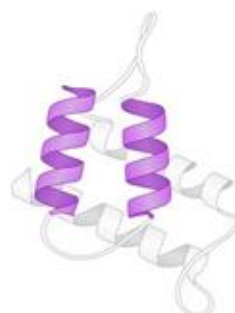
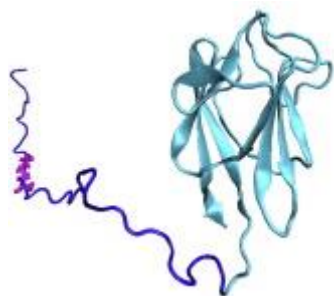
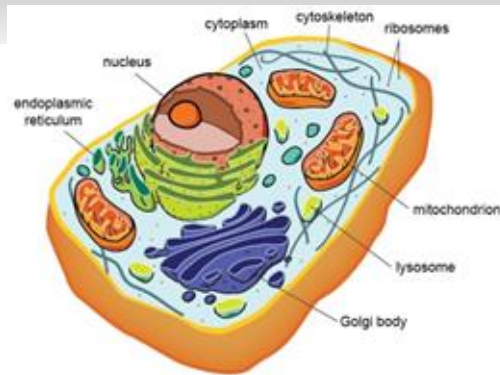
Strategies Compared in the Study

- Frozen $\leftrightarrow$ Lora
- Frozen $\leftrightarrow$ Full Fine Tuning

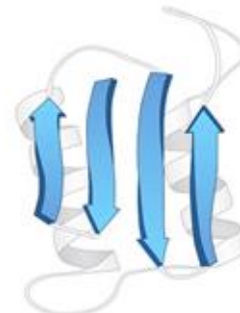


Datasets + downstream tasks

- Disorder
- Secondary Structure
- Subcellular location
- And more



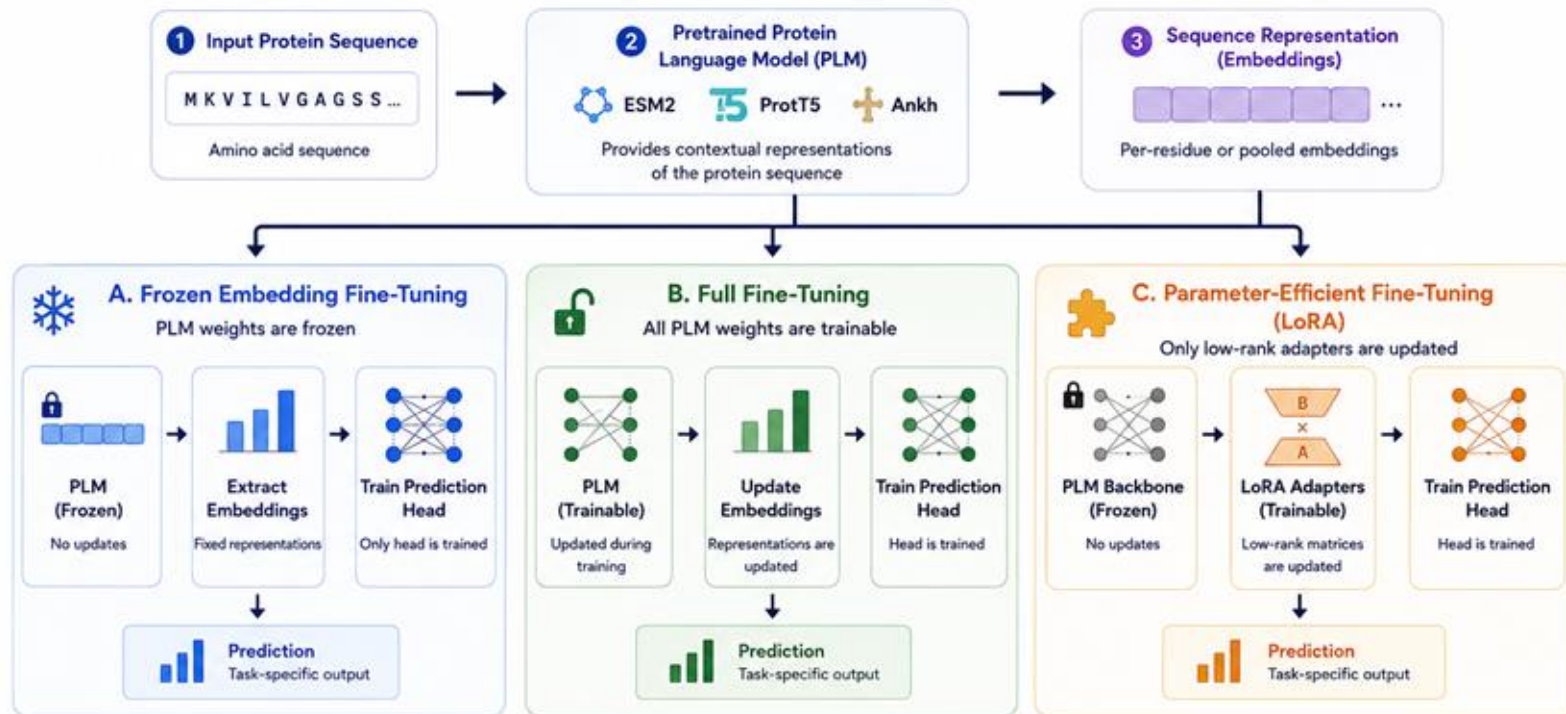
α - helices



β - pleated sheets



Workflow of Fine-Tuning Strategies for Protein Language Models



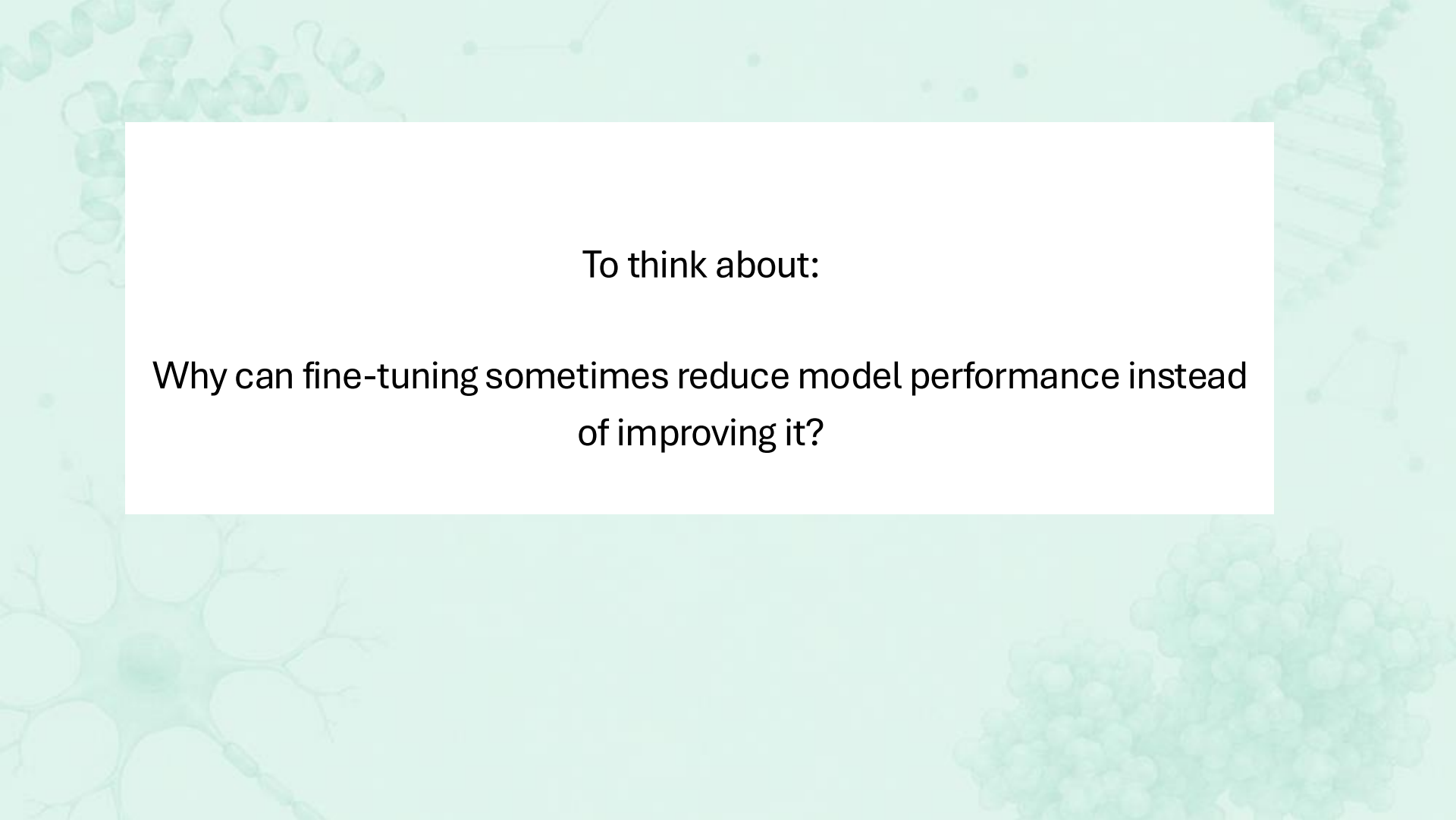


Results

- Fine-tuning protein language models boosts predictions across diverse tasks
- Difference is between frozen model, and fine-tuned task
- Asterisks (*) mark fully fine-tuned models;
- the others were LoRA-optimized
- LoRa does not perform much worse
- Best strategy is very down-stream task dependent

	GFP	AAV	GB1	Stability	Meltome	Subcellular location	Disorder	Secondary structure
ProtT5-	6.6±1.52	7.4±3.8	2.7±0.69	4.6±3.3	3.2±1.08	3.8±2.74	0.4±0.45	0.6±0.12
ESM2 8M*	4.9±0.5	15.8±3.11	6.5±1.36	1.5±3.98	2.0±0.86	3.9±2.11	1.8±0.79	0.8±0.18
ESM2 35M*	3.9±0.4	21.8±6.41	5.2±0.73	7.8±1.94	1.2±2.97	3.2±2.45	4.7±1.34	0.9±0.1
ESM2 150M*	5.2±0.33	18.9±2	4.6±1.14	-5.1±3.11	0.9±1.94	2.2±2.08	3.0±1.58	0.6±0.16
ESM2 650M*	4.0±0.51	31.2±5.04	2.2±1.31	7.9±7.56	0.8±2.03	0.9±2.57	0.0±0.78	0.8±0.08
ESM2 3B	4.9±0.22	8.1±1.21	2.7±0.99	5.2±1.12	2.2±1.55	3.6±1.4	0.6±0.97	0.8±0.1
Ankh base	3.2±0.35	17.6±5.22	1.8±1.62	5.3±6.17	3.5±1.95	2.2±1.31	-2.6±0.75	-0.4±0.08
Ankh large	2.5±0.49	11.7±2.84	3.4±1.68	11.3±6.64	-2.1±4.8	2.2±2.92	-1.1±0.8	-0.3±0.17
	mutational landscape			diverse dataset				

<https://doi.org/10.1038/s41467-024-51844-2>

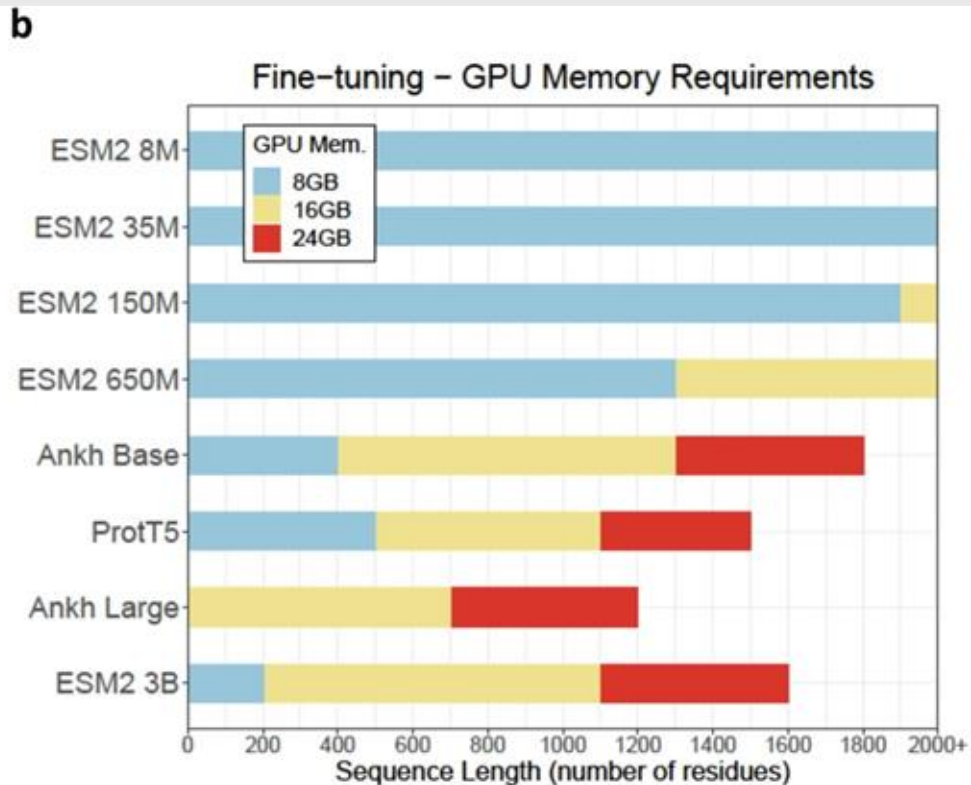
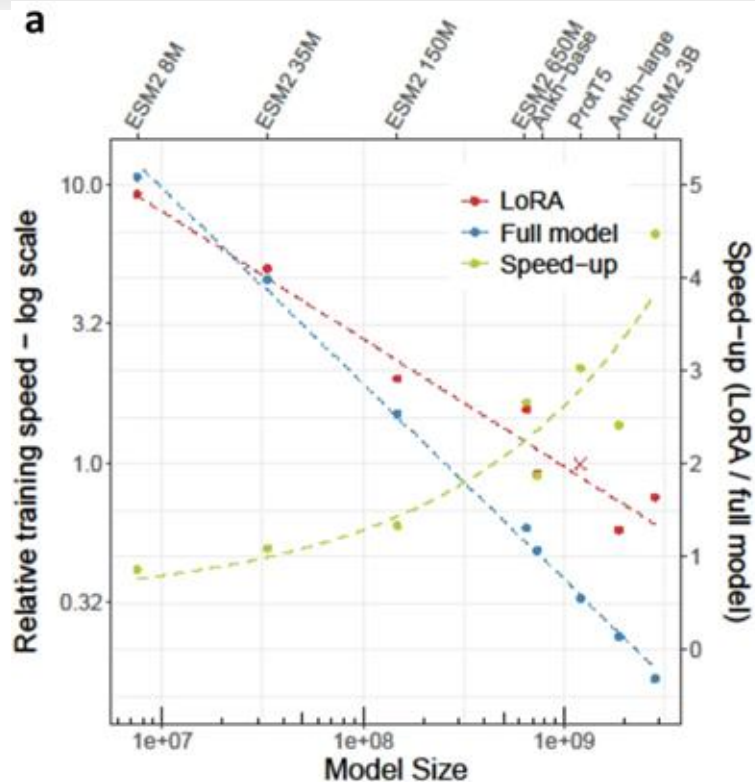


To think about:

Why can fine-tuning sometimes reduce model performance instead of improving it?

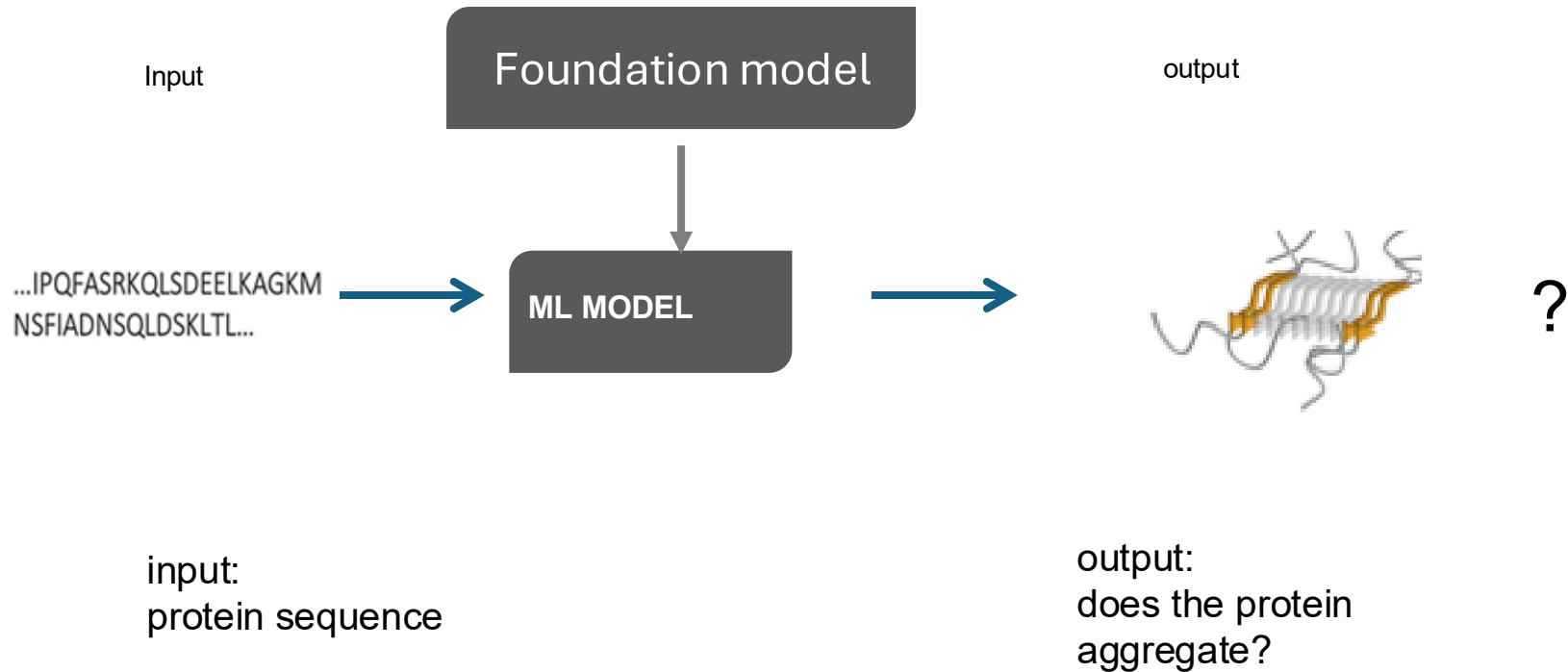


Strategies Compared in the Study



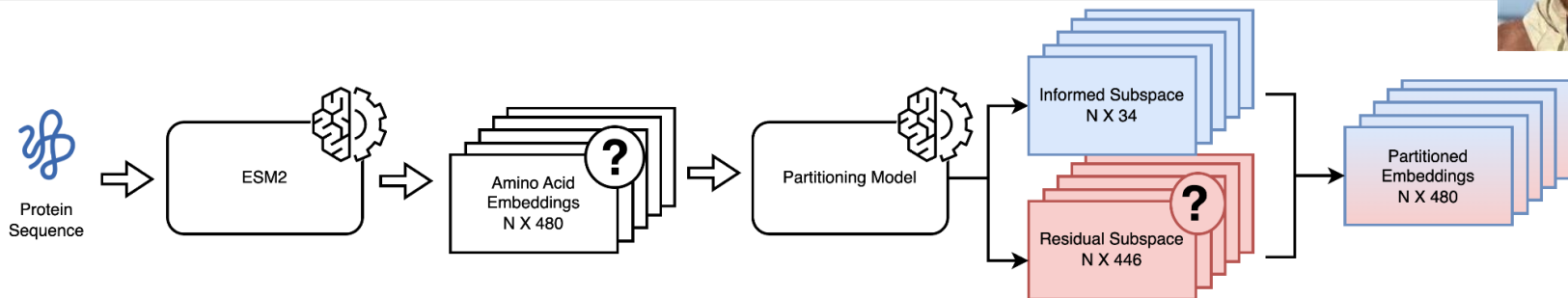


Explainable AI using foundation models?

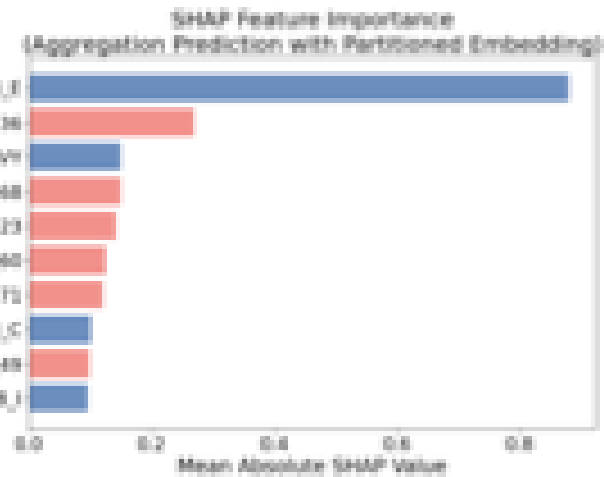




Explainable AI using protein foundation models

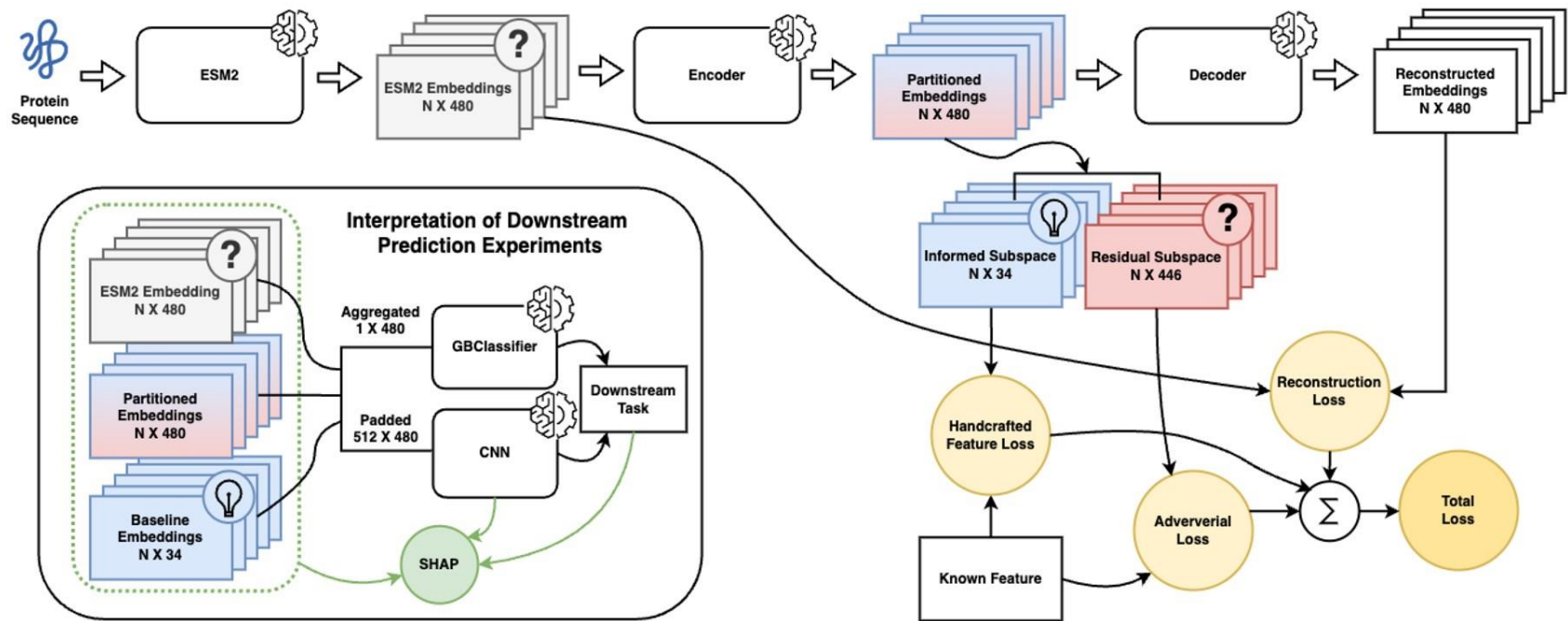


van Eck, J., Gogishvili, D., Silva, W., & Abeln, S. (2025).
 PLM-eXplain: Divide and Conquer the Protein Embedding Space.
arXiv preprint arXiv:2504.07156.



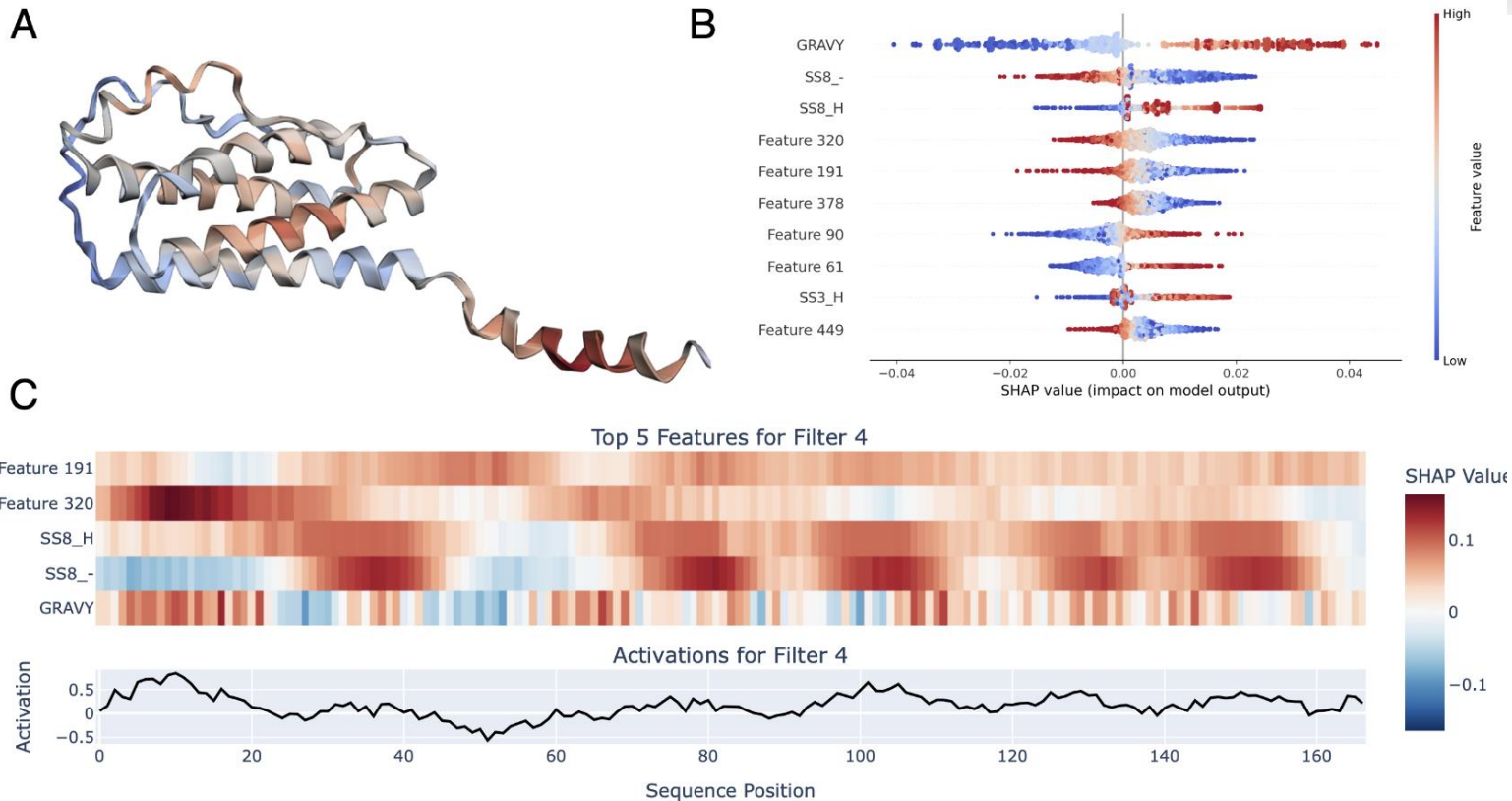


How to split the embeddings?





Detailed sequence-based explanations



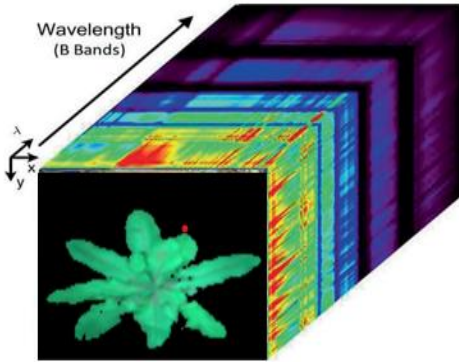


Between modalities: hyperspectral data and genotype

Frank Okyere

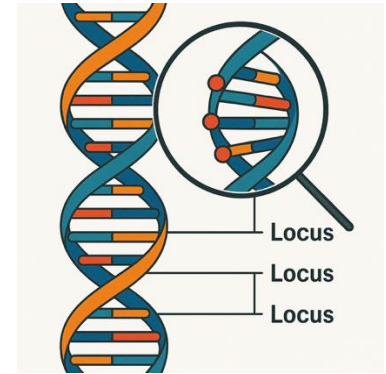


Input: hyper spectral data



machine learning
model

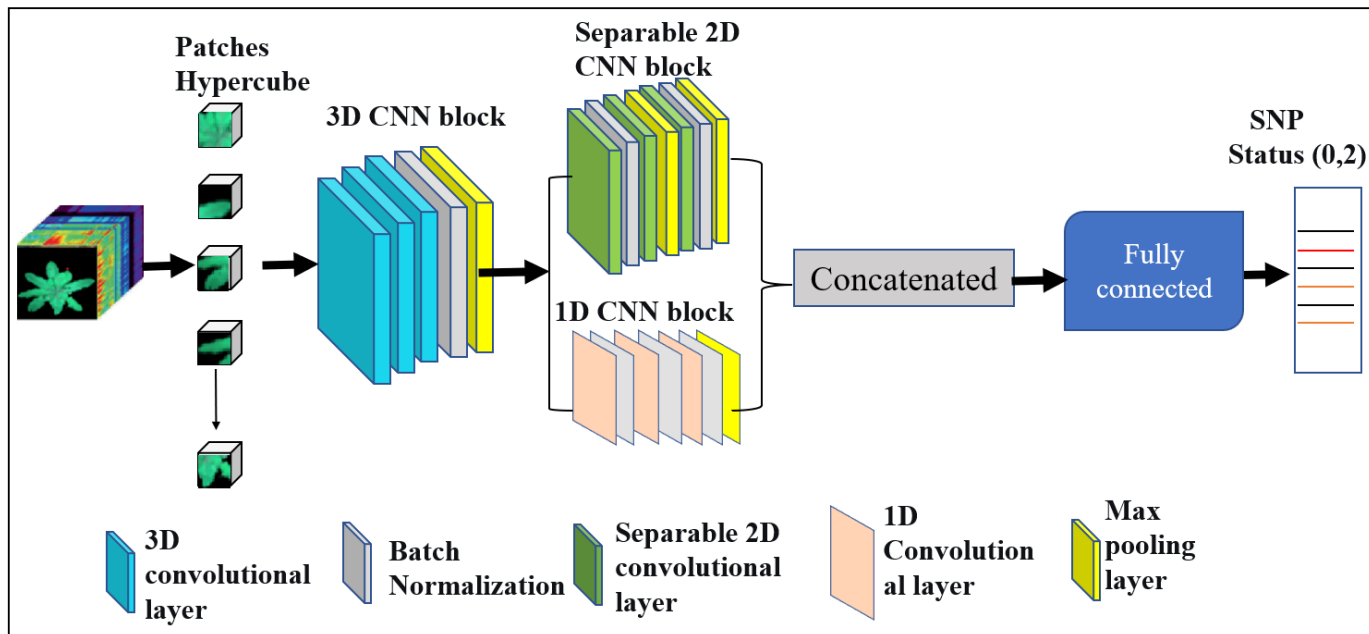
Output: genotype



We can learn complex relations between genotype and phenotype



Between modalities: hyperspectral data and genotype



We can learn complex relations between genotype and phenotype



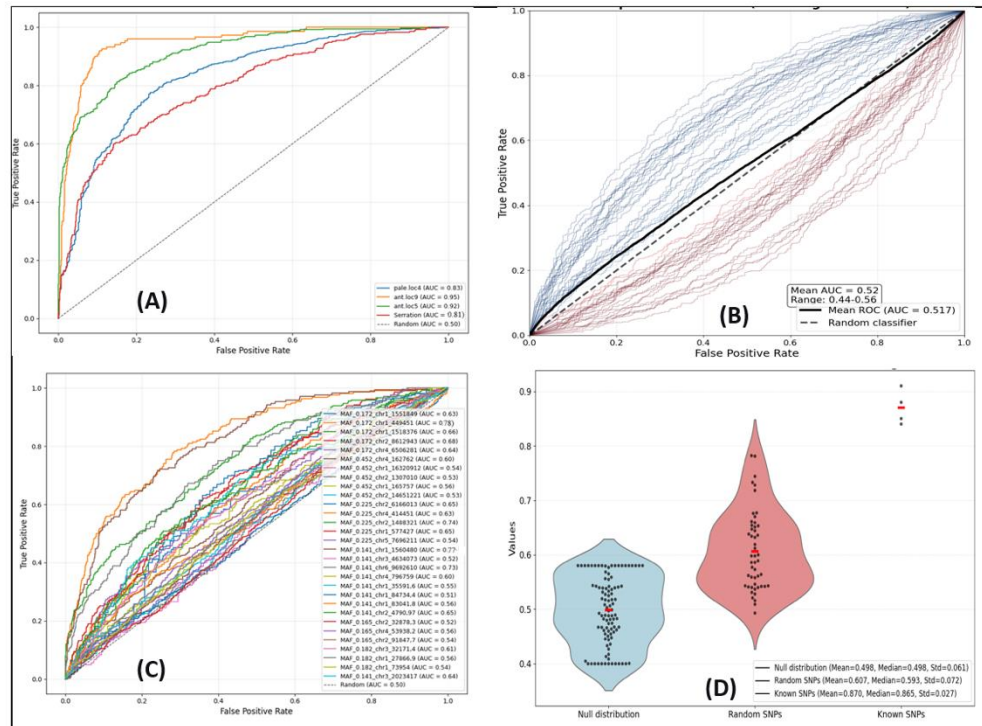
Between modalities: hyperspectral data and genotype

We can learn train the models, with relatively little annotated data;

We can predict that variant for SNP associated with

- Leaf ceration
- Anthoncyanin (purple color) production

Some uncharacterised SNPs also have an effect on the hyperstra

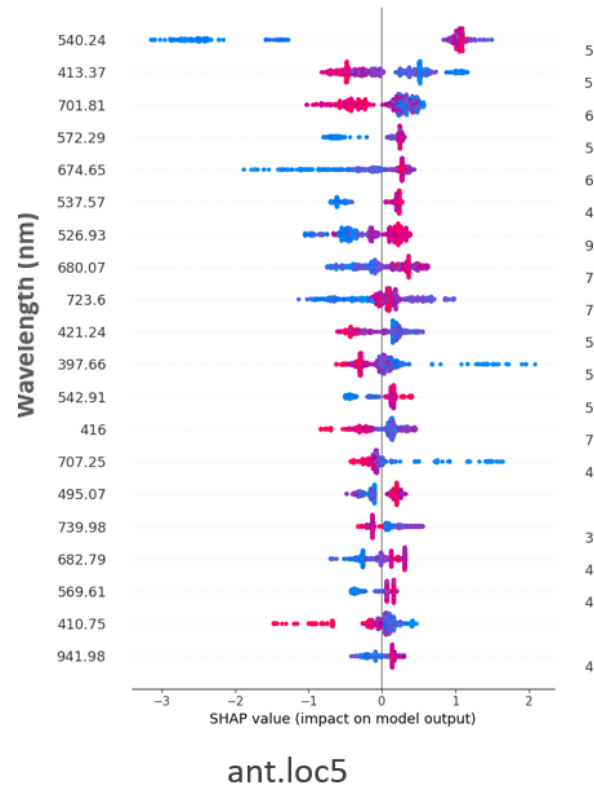
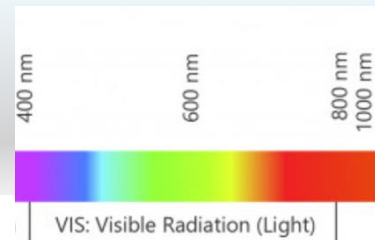
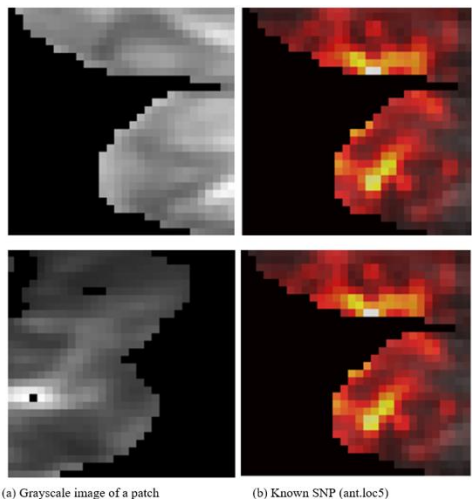




Between modalities: hyperspectral data and genotype

Explainable AI helps to
interpret the results.

Purple colour
waveband is important
for predictions.





Key points

- Life Science data provides specific challenges
- Protein language models, can help with a large number of downstream predictions tasks.
- There are different fine-tuning strategies, the best option is task dependent.
- We can obtain explainable models, even if we use foundation models.



AI Technology for Life group

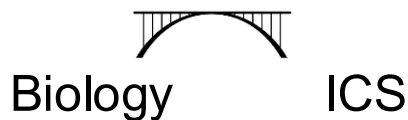


Utrecht
University

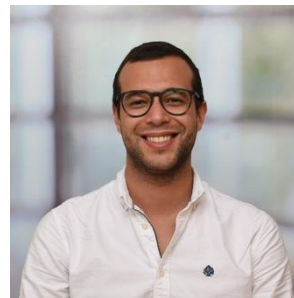
Ronnie de
Jonge



- AI for plant systems
- Plant – Microbiome



Wilson Silva



- Explainable AI
- Medical image analysis



Pavel Sinitcyn



- AI for proteomics
- De novo sequencing



Sanne Abeln



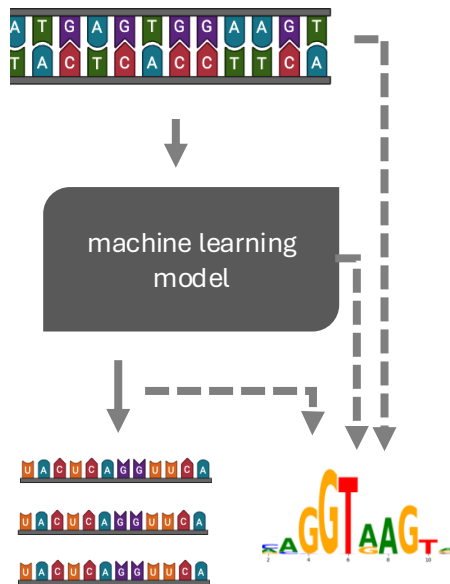
- Explainable AI for multi-omics
- simulations



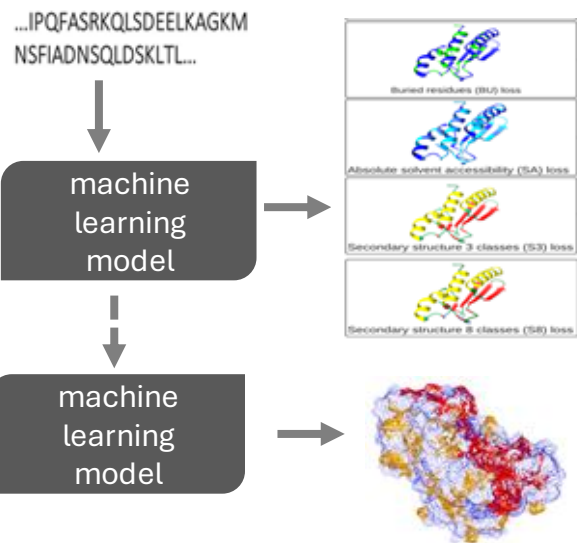


AI Technology for Life group: methodology development

Explainable AI



Transfer learning



Multi-modal data learning

